In [54]:

```
#Write a program to determine a given number is ‘odd’ or ‘even’ and
#print the following message “Number is ODD” or “Number is Even”.
n=int(input("Enter number: "))
if n%2==0:
    print("Number is Even")
else:
    print("Number is Odd")
```

```
Enter number: 26
Number is Even
```

In [55]:

```
#Write a program to check if the input number is positive, negative or zero
n=int(input("Enter number: "))
if n>0:
    print("Number is Positive")
elif n<0:
    print("Number is Negetive")
else:
    print("Number is zero")
```

```
Enter number: -23
Number is Negetive
```

In [62]:

```
#Write a program to find the maximum number among the three input numbers.
a=int(input("Enter first number: "))
b=int(input("Enter second number: "))
c=int(input("Enter third number: "))
if a>b and a>c:
    print(a,"is maximum")
elif b>c and b>a:
    print(b,"is maximum")
else:
    print(c,"is maximum")
```

```
Enter first number: 2
Enter second number: 5
Enter third number: 3
5 is maximum
```

In [38]:

```
#Write a program to check if year is a leap year or not (Nested If)
year=int(input("Enter year: "))
if year%4 == 0:
    if year%100 !=0 or year%400==0:
        print(year,"is a leap year")
    else:
        print(year,"is not a leap year")
else:
        print(year,"is not a leap year")
```

```
Enter year: 2020
2020 is a leap year
```

In [39]:

```
#Write a program to find sum of first N natural number given by user.
n=int(input("Enter N natural number: "))
sum=0
for i in range(n+1):
    sum += i
print(sum)
```

```
Enter N natural number: 21
231
```

In [40]:

```
#Write a program to find average of first N natural number given by user.
n=int(input("Enter N natural number: "))
sum=0
count=0
for i in range(n+1):
    sum += i
    count+=1
print(sum/count)
```

```
Enter N natural number: 16
8.0
```

In [41]:

```
#Write a python program to read three numbers (a,b,c) and check how many numbers between ‘a
a=int(input("Enter first number: "))
b=int(input("Enter second number: "))
c=int(input("Enter number to check divisibility: "))
count=1
for i in range(a,b+1):
    if i%c==0:
        count += 1
print(f"the count of numbers between {a} and {b}, divisible by {c} are {count}" )
```

```
Enter first number: 2
Enter second number: 36
Enter number to check divisibility: 3
the count of numbers between 2 and 36, divisible by 3 are 13
```

In [42]:

```
#Write a Python program that prints all the numbers from 0 to 6 except 3 and 6.
for i in range (7):
    if i==3 or i==6:
        continue
    else:
        print(i)
```

```
0
1
2
4
5
```

In [43]:

```
#Write a Python program to print the multiplication table of given number by user
a=int(input("Enter number for table: "))
for i in range(1,11):
    print(f"{a}*{i}={a*i}")
```

```
Enter number for table: 17
17*1=17
17*2=34
17*3=51
17*4=68
17*5=85
17*6=102
17*7=119
17*8=136
17*9=153
17*10=170
```

In [45]:

```
#Write a program to find the factorial of a number provided by the user
a=int(input("Enter number: "))
fact=1
for i in range(1,a+1):
    fact *= i
print("factorial of",a,"is",fact)
```

```
Enter number: 6
factorial of 6 is 720
```

In [48]:

```
# Write a python program to display the Fibonacci sequence up to n-th term
nterms = int(input("How many terms? "))

n1, n2 = 1, 1
count = 0
if nterms <= 0:
   print("Please enter a positive integer")
elif nterms == 1:
   print("Fibonacci sequence upto",nterms,":")
   print(n1)
else:
   print("Fibonacci sequence:")
   while count < nterms:
       print(n1,end=" ")
       nth = n1 + n2
       n1 = n2
       n2 = nth
       count += 1
```

```
How many terms? 10
Fibonacci sequence:
1 1 2 3 5 8 13 21 34 55
```

In [49]:

```
#Write a program to take 10 values from keyboard using loop and print their average on the
sum=0
count=0
for i in range(10):
    a=int(input("Enter number: "))
    sum=sum+a
    count+=1
print(sum/count)
```

```
Enter number: 2
Enter number: 1
Enter number: 3
Enter number: 5
Enter number: 4
Enter number: 6
Enter number: 7
Enter number: 8
Enter number: 9
Enter number: 10
5.5
```

In [50]:

```
#Write a program to reverse a number.
a=int(input("Enter number: "))
sum=0
while a!=0:
    r=a%10
    sum=sum*10+r
    a=a//10
print("reversed number is",sum)
```

```
Enter number: 2156
reversed number is 6512
```

In [51]:

```
#Write a program to check whether a number is Armstrong number or not
a=int(input("Enter number: "))
temp=a
count=0
while a!=0:
    a//=10
    count+=1
a=temp
sum=0
while a!=0:
    r=a%10
    sum=sum+r**count
    a//=10
if temp==sum:
    print("Number is Armstrong")
else:
    print("Number is not Armstrong")
```

```
Enter number: 1634
Number is Armstrong
```

In [52]:

```
# Write a program to check if a number is prime or not.
a=int(input("Enter number: "))
flag=0
for i in range(2,a):
    if a%i==0:
        flag=1
        break
if flag==0:
    print("Prime Number")
else:
    print("Not a Prime Number")
```

```
Enter number: 17
Prime Number
```

In [53]:

```
# Write a program to print prime number between given interval from user
a=int(input("Enter 1st number: "))
b=int(input("Enter 2nd number: "))
for i in range(a,b+1):
    flag=0
    for j in range(2,i):
        if i%j==0:
            flag=1
            break
    if flag==0:
        print(i)
```

```
Enter 1st number: 3
Enter 2nd number: 50
3
5
7
11
13
17
19
23
29
31
37
41
43
47
```

In [10]:

```
n=int(input("Enter no. of rows: "))
for i in range(n):
    for j in range(i+1):
        print("*",end=" ")
    print()
```

```
Enter no. of rows: 5
*
* *
* * *
* * * *
* * * * *
```

In [7]:

```
n=int(input("Enter no. of rows: "))
for i in range(n):
    for j in range(i):
        print(" ",end=" ")
    for k in range(n,i,-1):
        print("*",end=" ")
    print()
```

```
Enter no. of rows: 5
* * * * *
  * * * *
    * * *
      * *
        *
```

In [8]:

```
n=int(input("Enter no. of rows: "))
for i in range(n):
    for j in range(i):
        print(" ",end="")
    for k in range(n,i,-1):
        print("*",end=" ")
    print()
```

```
Enter no. of rows: 5
* * * * *
 * * * *
  * * *
   * *
    *
```

In [9]:

```
n=int(input("Enter no. of rows: "))
for i in range(n):
    for j in range(n,i+1,-1):
        print(" ",end=" ")
    for k in range(i+1):
        print("*",end=" ")
    print()
```

```
Enter no. of rows: 5
        *
      * *
    * * *
  * * * *
* * * * *
```

In [11]:

```
n=int(input("Enter no. of rows: "))
for i in range(n):
    for j in range(n,i+1,-1):
        print(" ",end="")
    for k in range(i+1):
        print("*",end=" ")
    print()
```

```
Enter no. of rows: 5
    *
   * *
  * * *
 * * * *
* * * * *
```

In [12]:

```
n=int(input("Enter no. of rows: "))
for i in range(n):
    for j in range(n,i,-1):
        print("*",end=" ")
    print()
```

```
Enter no. of rows: 5
* * * * *
* * * *
* * *
* *
*
```

In [17]:

```
n=int(input("Enter no. of rows: "))
for i in range(n+1):
    for j in range(1,n+1-i):
        print(j,end=" ")
    print()
```

```
Enter no. of rows: 5
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

In [18]:

```
n=int(input("Enter no. of rows: "))
for i in range(n+1):
    for j in range(1,i+1):
        print(j,end=" ")
    print()
```

```
Enter no. of rows: 5

1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

In [19]:

```
n=int(input("Enter no. of rows: "))
for i in range(n+1):
    for j in range(1,i+1):
        print(i,end=" ")
    print()
```

```
Enter no. of rows: 5

1
2 2
3 3 3
4 4 4 4
5 5 5 5 5
```

In [20]:

```
n=int(input("Enter no. of rows: "))
for i in range(n):
    for j in range(i+1):
        if i%2==0:
            print("*",end=" ")
        else:
            print("#",end=" ")
    print()
```

```
Enter no. of rows: 4
*
# #
* * *
# # # #
```

In [26]:

```
n=int(input("Enter no. of rows: "))
for i in range(n):
    for j in range(i+1):
        print((i+j+1)%2,end=" ")
    print()
```

```
Enter no. of rows: 5
1
0 1
1 0 1
0 1 0 1
1 0 1 0 1
```

In [4]:

```
n=int(input("Enter no. of rows: "))
for i in range(n+1):
    for k in range(n,i,-1):
        print("",end=" ")
    for j in range(1,i+1):
        print(j,end=" ")
    print()
```

```
Enter no. of rows: 5

    1
   1 2
  1 2 3
 1 2 3 4
1 2 3 4 5
```

In [5]:

```
n=int(input("Enter no. of rows: "))
for i in range(n+1):
    for k in range(n,i,-1):
        print("",end=" ")
    for j in range(1,i+1):
        print(i,end=" ")
    print()
```

```
Enter no. of rows: 5

    1
   2 2
  3 3 3
 4 4 4 4
5 5 5 5 5
```

In [6]:

```
n=int(input("Enter no. of rows: "))
for i in range(n):
    for k in range(n,i,-1):
        print("",end=" ")
    for j in range(i+1):
        if i%2==0:
            print("*",end=" ")
        else:
            print("#",end=" ")
    print()
```

```
Enter no. of rows: 4
    *
   # #
  * * *
 # # # #
```